# Wooden Tower Project Summary Report

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## 1. Project Overview

I designed and built a 7.3-gram, three-legged balsa tower that held 14,500 g and placed first at my school's 2024 Science Olympiad selection and at the West Suburban Science League Wooden Tower Event.

| Item | Details |
|---|---|
| Events | Massachusetts Science Olympiad, Tower (school selection);<br>West Suburban Science League (WSSL), Wooden Tower |
| Year | 2024 |
| Division | C |
| Team | Individual |
| My role | Design, construction, testing |
| Time spent | 8 weeks, 3 towers built |
| Results | 1st place at school selection, about 400 points ahead of 2nd place;<br>1st place at WSSL |

## 2. Challenge and Constraints

The goal was the highest structural efficiency: the most load held for the least tower mass.

Score = (Load held + Bonus) ÷ Tower mass, multiplied by 1.25 for a compliant design log. A 5,000 g bonus applied only to towers that held the full 15 kg while standing outside a 29 cm circle on the test base.

- Minimum height: 60 cm (to the loading block)
- Base opening the tower had to span: 20 cm × 20 cm
- Maximum test load: 15,000 g

- Loading Block: 5 cm × 5 cm, sitting on top of the tower, with a chain running down through the tower to a bucket of sand
- Material choice: I built with balsa because it's very light, but it's soft and brittle, so my design had to work around how easily it breaks.

## 3. Initial Design

I chose a three-legged tower. A triangle is the only shape that can't deform without one of its sides changing length, so a three-sided tower stays rigid with less bracing than a four-sided one. Three legs also mean less mass than four, and three points always sit flat on the test base, so the tower doesn't rock under load.

- Legs: 1/8" × 1/8" balsa
- Braces: 1/16" × 1/16" balsa
- Glue: Gorilla Super Glue (cyanoacrylate)
- Bracing pattern for my winning tower: Z-bracing. On each face, the braces zigzag between the two legs. On my third tower, every brace joins the legs at a 90-degree angle.
- Design Tool: Paper sketches, a 3D model in Blender, and a 3D printed jig to hold the legs and braces in position while gluing.

## 4. Prototypes and Testing

All three towers failed the same way: a main leg bowed outward near the base. In my first two towers, I hadn't yet set the braces at 90 degrees to the legs, so the legs had too little support against bending sideways. My third tower still failed near the base, but after I rejoined the braces on all three main legs at 90 degrees and added a few braces near the bottom, it held much more weight.

| Prototype | Load Held (g) | Mass (g) | How it failed | Where it failed |
|---|---|---|---|---|
| 1 and 2 | 9,500 | 5.0 | Main leg bowed outward | Near the base |
| 3 | 14,500 | 7.3 | Main leg bowed outward | Near the base |

## 5. Failure Analysis

In my first two towers, a leg failed about 0.5 cm above the table, in the bottom segment between the last brace and the ground, where nothing supported it. The leg didn't crush; it buckled. A slender column under compression stays straight until the load reaches a

critical value, and then it suddenly bends sideways. That critical load is given by Euler's formula:

$$P = \pi^2 EI / (KL)^2$$

Here E is the stiffness of the wood, I describes how well the leg's cross-section resists bending, L is the length of the leg between braces, and K is a factor that depends on how the ends are held. Because the critical load falls with the square of L, the longest unbraced segment fails first, and in my towers those segments were near the base. The legs also lean inward, so the load pushes their lower ends outward, which adds to the bowing. The leg that failed had also been slightly chipped by a miscut, which made an already weak spot even weaker.

At first, I didn't understand why the legs failed, and I didn't even know this equation existed. Only after researching the problem did I find it, and as I dug into the reasoning behind it, I began to see how it explained the way the legs of my tower compressed and bent.

## 6. Key Design Change: 90-degree Brace Joints and Added Braces

For the third tower, I made two changes. First, I rejoined every brace on all three legs, from top to bottom, so that each brace meets the leg at 90 degrees. Second, I added a few braces near the bottom to close the unbraced gap where my earlier towers had failed.

- **Why the bottom braces:** The leg that failed had been slightly chipped by a miscut, and its lowest section, between the last brace and the table, had no bracing to protect it.
- **Why the changes increased the load:** Since the legs failed by buckling rather than crushing, shortening the unbraced length at the bottom made the biggest difference, because the load a leg can carry rises sharply as that length shrinks, by the square. Meeting the leg at 90 degrees lets each brace hold the leg firmly in place at that point, and the braces also act as ties, pulling against the outward push of the tapered legs so the base cannot spread.
- **Mass and load:** The changes added about 2.3 grams, but the tower held 14,500 grams instead of 9,500, an increase of 5,000 grams. Its structural efficiency, the load held divided by the tower's own mass, rose from 1,900 to about 1,986.

## 7. Conclusion

My first two towers buckled at the same weak point near the base. After learning why, I rejoined the braces on all three main legs at 90 degrees and added braces near the bottom. My third tower held 14,500 grams, and its efficiency rose from 1,900 to about 1,986. This project taught me to treat each failure as data.

**Figure 1.** Top view of the 3D model

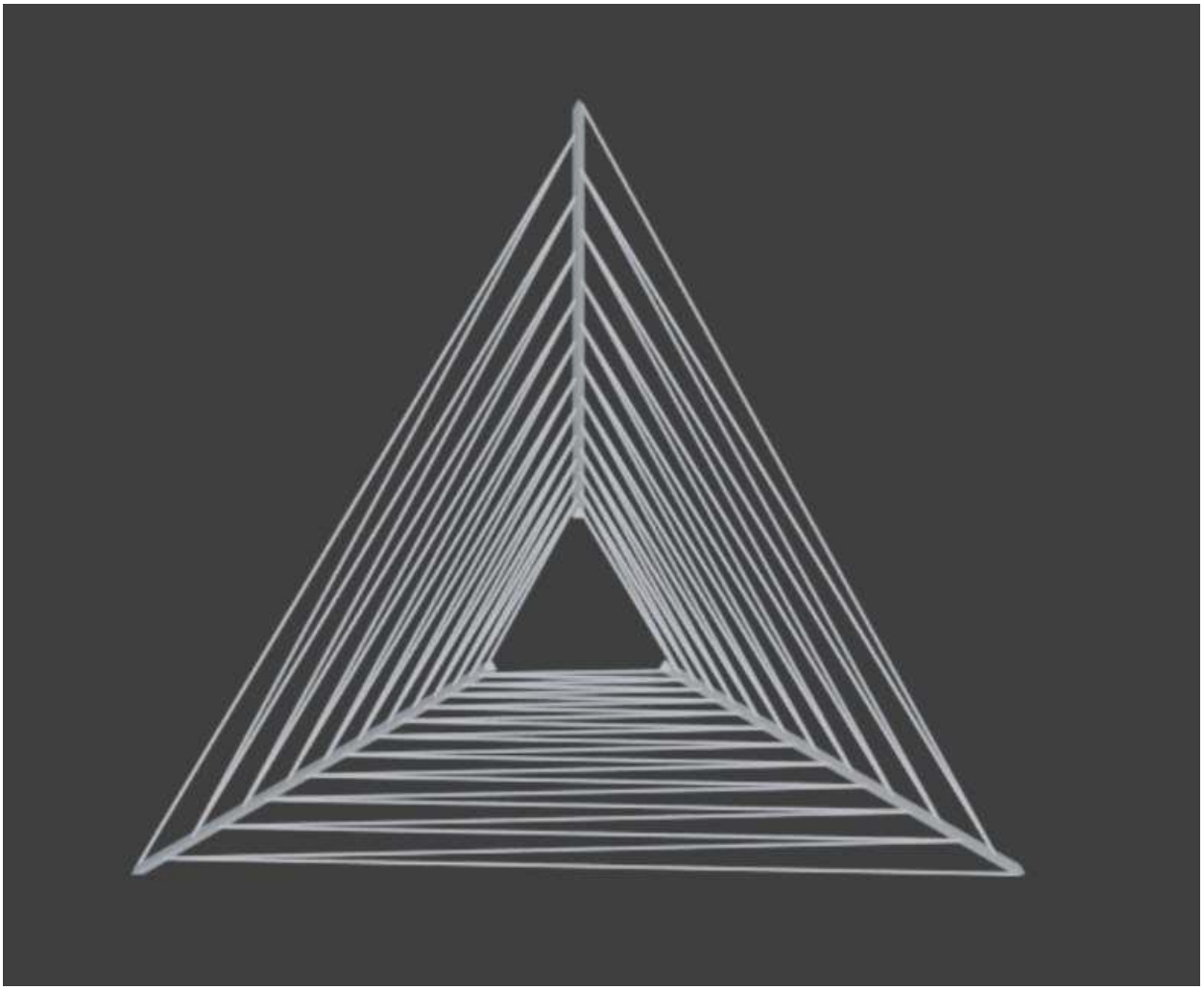


**Figure 2.** Bottom view of the 3D model

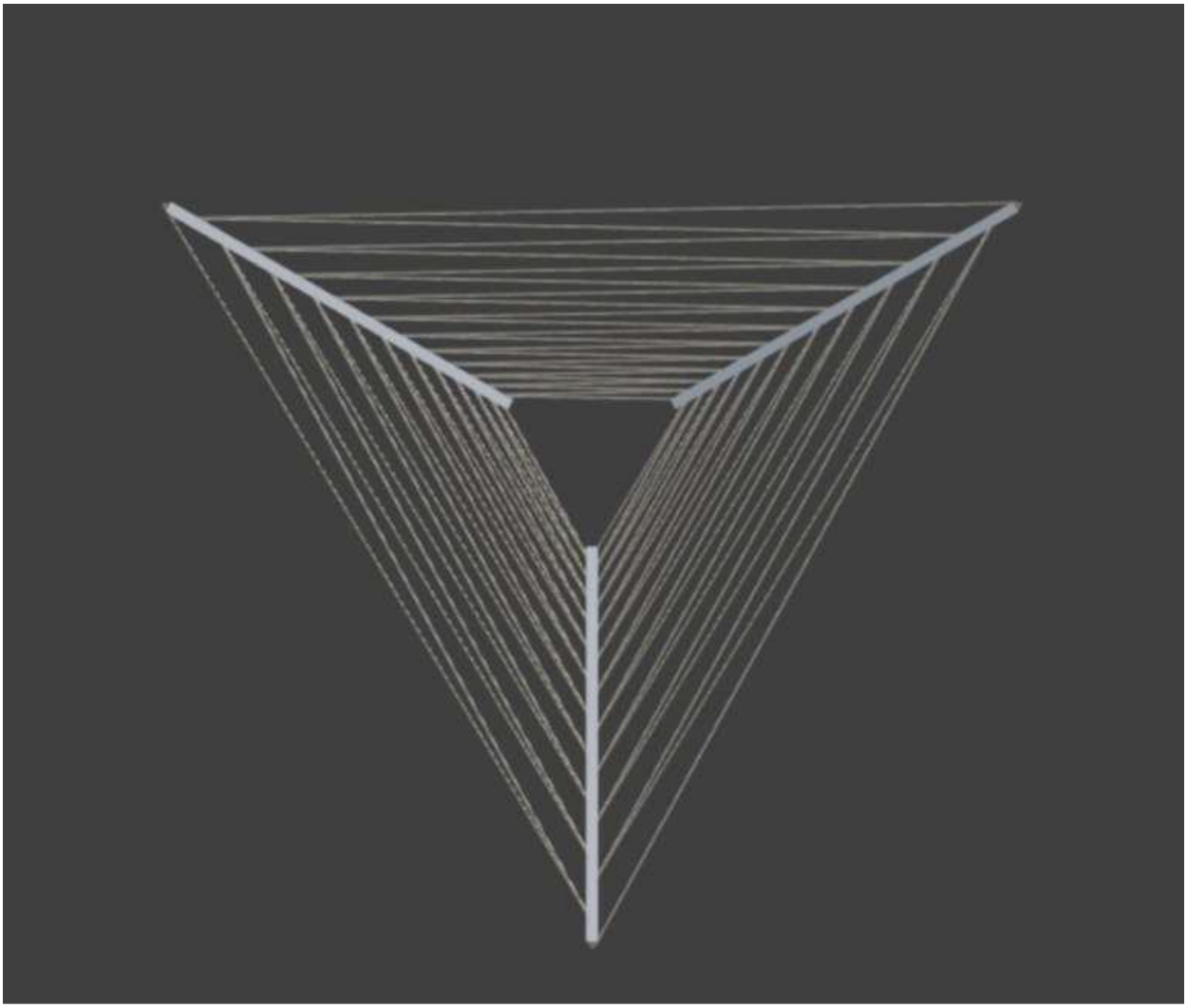


**Figure 3.** Side view of the 3D model, with the 90-degree brace-leg joints.

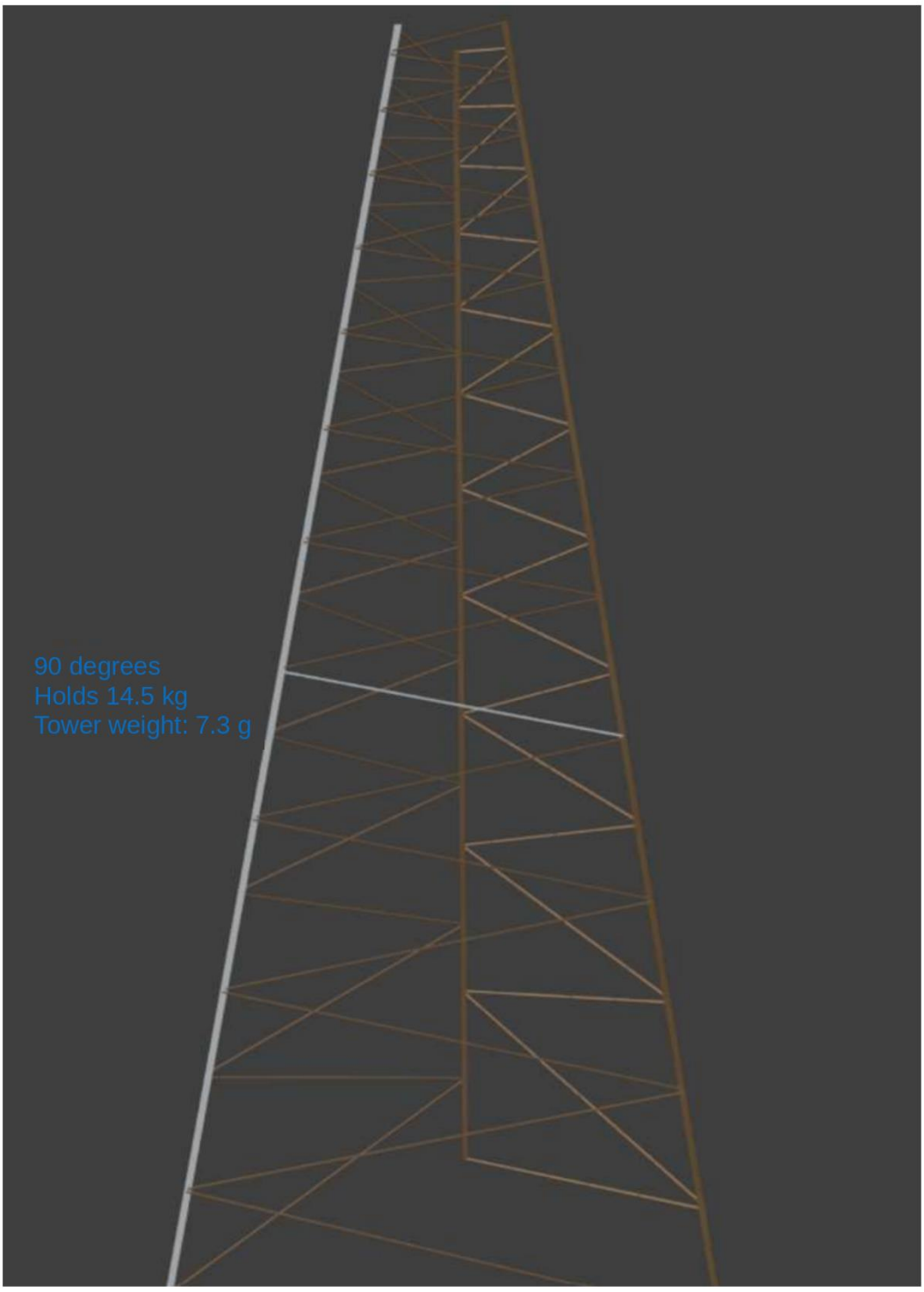


**Figure 4.** 3D model of the finished tower

